OBJECTIVE WORKSHEET

Time and Motion

Science — Physics · Class 7 · Max Marks: 25 · Time: 45 min

Section A — Multiple Choice Questions (1 × 8 = 8)

1. A car travels 240 km in 4 hours. Its average speed is:

A. 4 km/h
B. 240 km/h
C. 96 km/h
D. 60 km/h

2. The SI unit of speed is:

A. metre
B. m/s
C. second
D. km/h

3. A speed of 72 km/h is equal to:

A. 20 m/s
B. 72 m/s
C. 7.2 m/s
D. 200 m/s

4. The time taken by a pendulum to complete one full oscillation is called its:

A. amplitude
B. frequency
C. time period
D. speed

5. Which of these is an example of periodic motion?

A. a stone falling from a roof
B. a car moving on a straight road
C. a bus starting off
D. the pendulum of a wall clock

6. On a distance–time graph, a horizontal line (parallel to the time axis) shows that the object is:

A. at rest
B. moving slowly
C. speeding up
D. moving fast

7. A body that covers equal distances in equal intervals of time is said to be in:

A. circular motion
B. non-uniform motion
C. uniform motion
D. no motion

8. The spinning blades of a moving fan show:

A. rectilinear motion
B. rotational (circular) motion
C. no motion
D. straight-line motion

Section B — Fill in the Blanks (1 × 5 = 5)

1. Speed = distance ÷ ______.

2. The device in a vehicle that measures its speed is the ______.

3. The device that records the total distance travelled by a vehicle is the ______.

4. One complete to-and-fro movement of a pendulum is called one ______.

5. If a body covers unequal distances in equal intervals of time, its motion is ______.

Section C — True or False (1 × 5 = 5)

1. A body moving with uniform speed covers equal distances in equal intervals of time. (True / False)

2. km/h is the SI unit of speed. (True / False)

3. The time period of a simple pendulum depends on its length. (True / False)

4. On a distance–time graph, a steeper line means a slower speed. (True / False)

5. The motion of a child on a swing is periodic. (True / False)

Section D — Match the Columns (1 × 4 = 4)

Column A	Column B
1. Speedometer	a) measures the distance travelled
2. Odometer	b) SI unit of speed
3. Pendulum clock	c) measures speed
4. metre/second	d) uses periodic motion to keep time

Section E — Assertion–Reason (1 × 3 = 3)

Choose: (A) both A and R true and R explains A; (B) both true but R does not explain A; (C) A true, R false; (D) A false, R true.

1. **Assertion (A):** A pendulum clock can be used to measure time.
   Reason (R): A given simple pendulum takes the same time for each oscillation.
2. **Assertion (A):** On a distance–time graph, a straight slanting line shows uniform speed.
   Reason (R): Equal distances are covered in equal intervals of time.
3. **Assertion (A):** A speed of 36 km/h is slower than a speed of 36 m/s.
   Reason (R): 1 m/s is greater than 1 km/h.

Answer Key

Multiple Choice Questions

1. D	**2.** B	**3.** A	**4.** C	**5.** D
6. A	**7.** C	**8.** B		

Fill in the Blanks

1. time
2. speedometer
3. odometer
4. oscillation
5. non-uniform

True or False

1. True
2. False
3. True
4. False
5. True

Match the Columns

1-c, 2-a, 3-d, 4-b

Assertion–Reason

1. A
2. A
3. A